

University of Amsterdam



MLRC 2021 - Our Experience

Authors: Piyush Bagad, Danilo de Goede, Paul Hilders, Jesse Maas

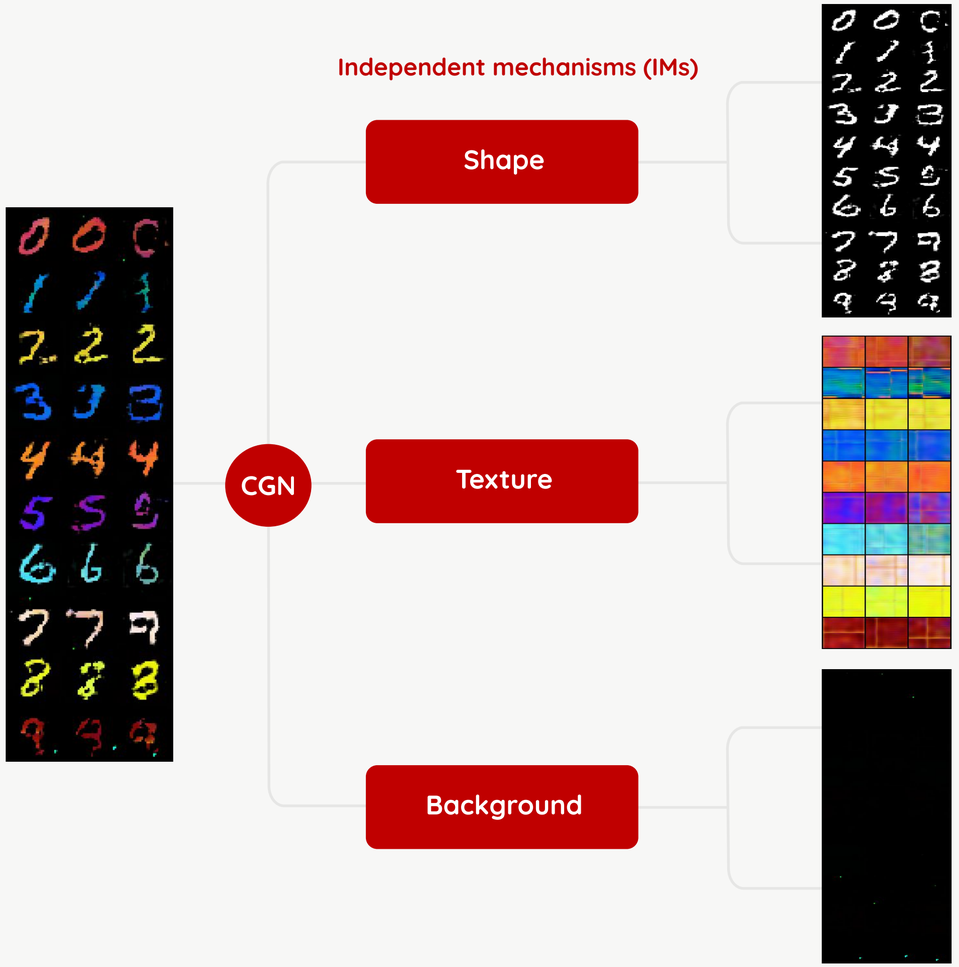
Supervisor: Christos Athanasiadis

Talk Overview

- Our work from FACT 2022
  - Context
  - The Counterfactual Generative Network (CGN)
  - Scope of Reproducibility
  - Our methodology and Results
- Lessons learnt, Tips and Suggestions

Context

- Deep Learning models tend to learn “shortcuts” that perform well on benchmarks.
- Shortcut learning causes models to be more sensitive to input perturbation and unseen input contexts.
- Sauer and Geiger (2021) propose an approach using a Counterfactual Generative Network.



Counterfactual Generative Network

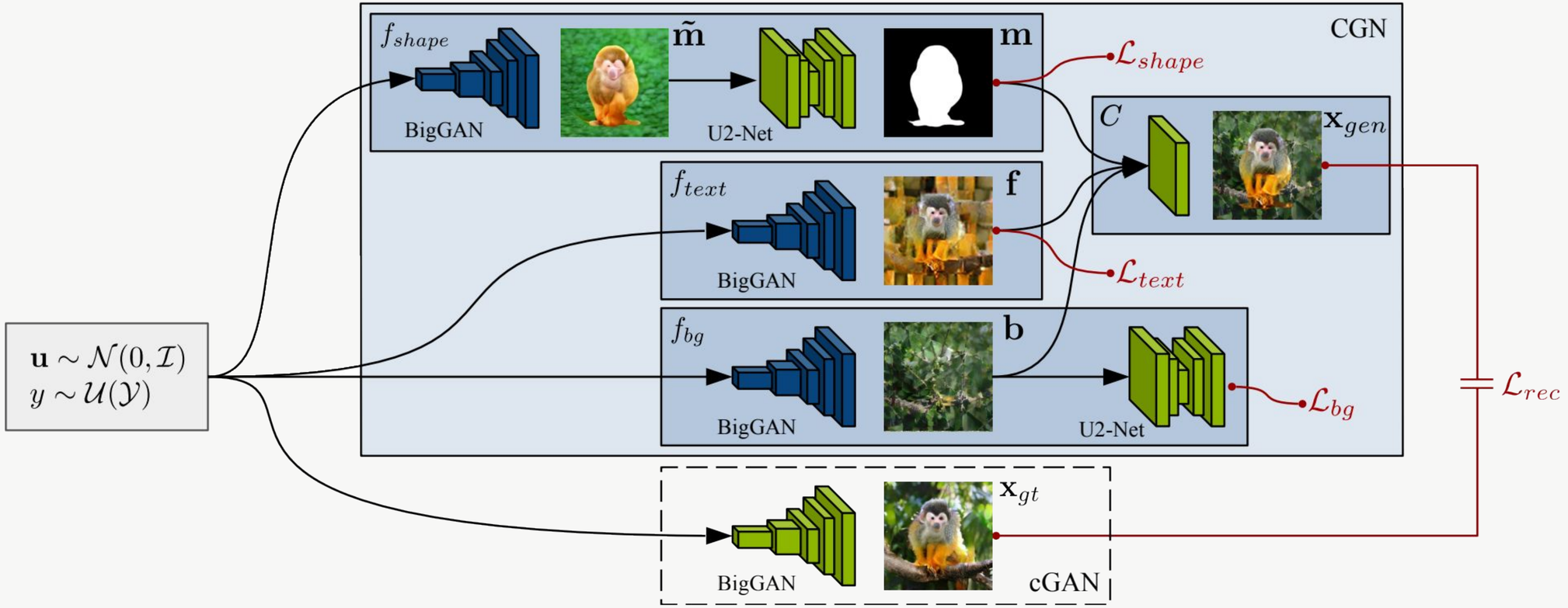


Figure 1. Architecture overview (ImageNet) of the Counterfactual Generative Network (Sauer and Geiger, 2021)

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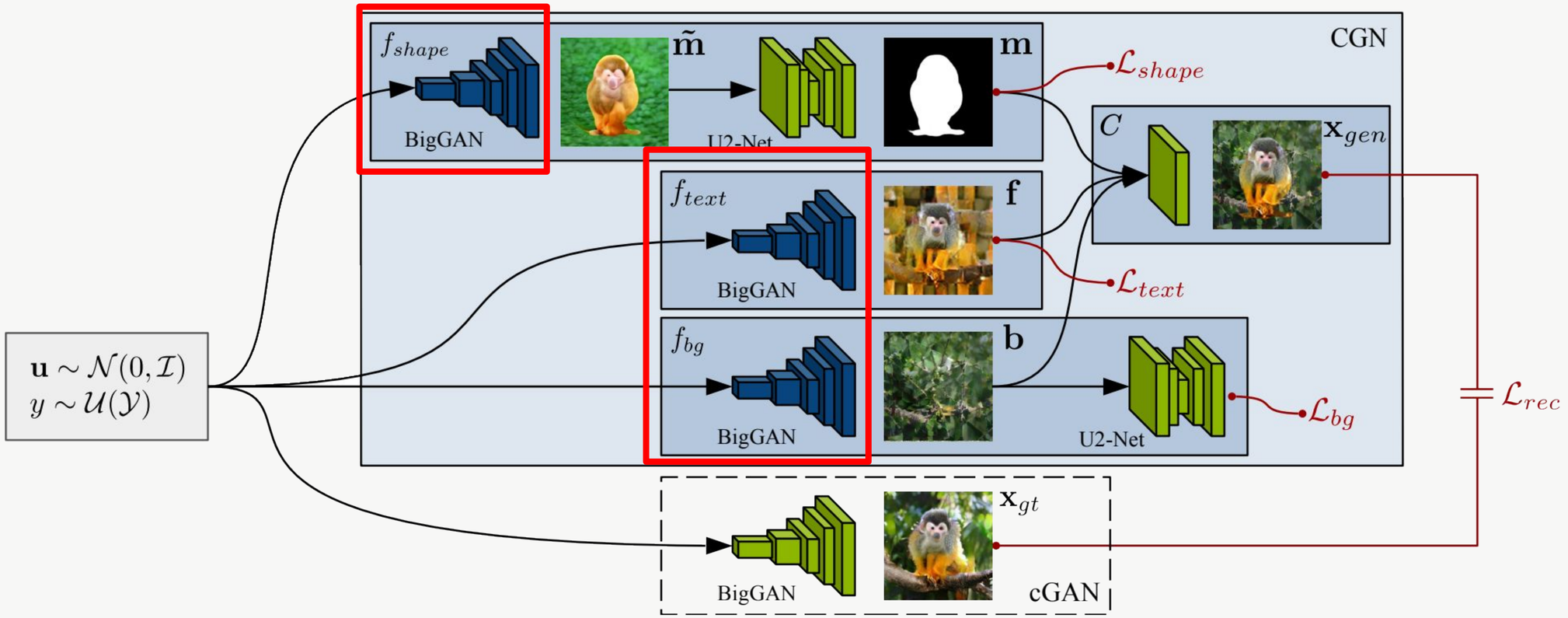


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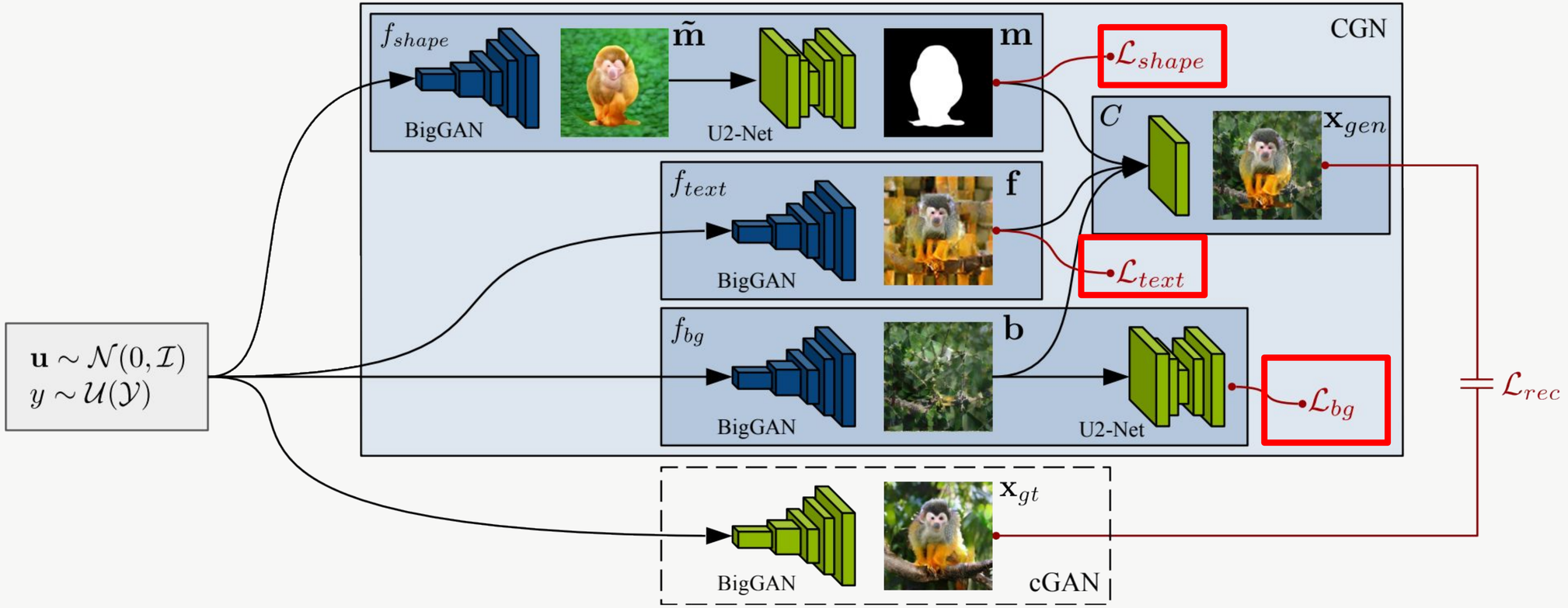


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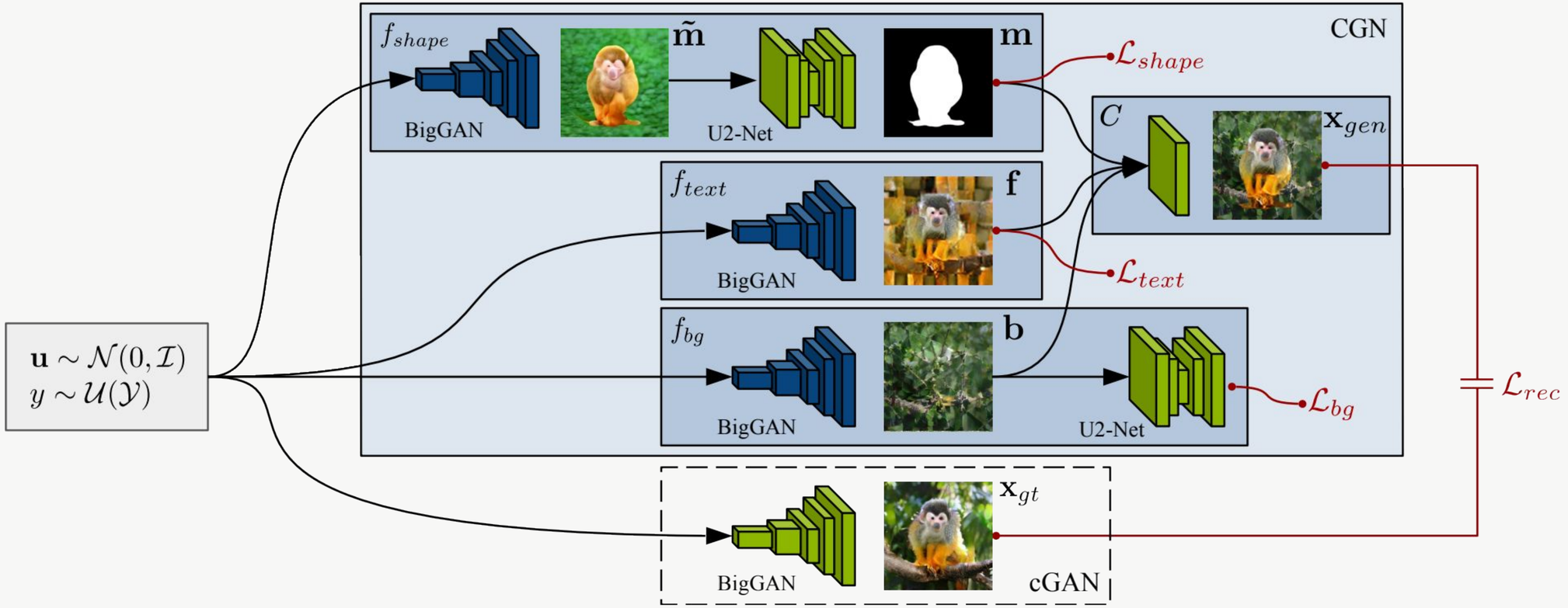
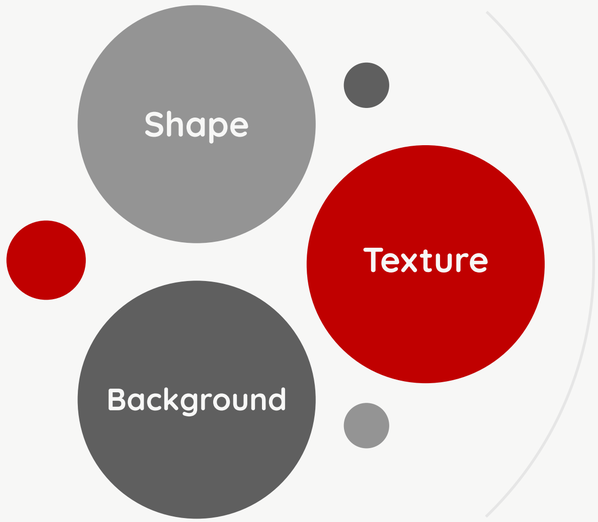


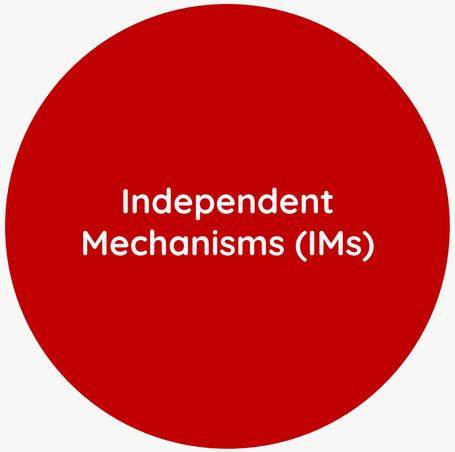
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Scope of Reproducibility

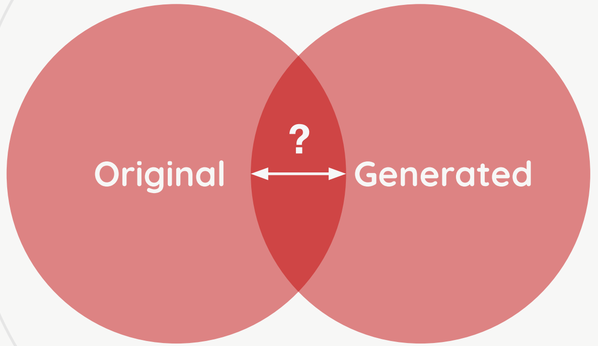
Main claims of the original paper



High-Quality Counterfactuals (HQC)



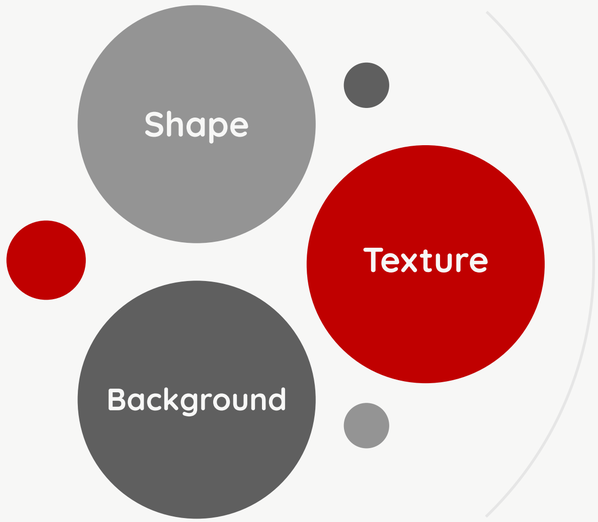
Inductive Bias Requirements (IBR)



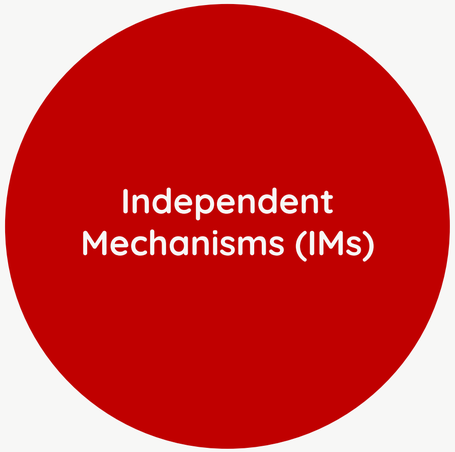
Out-of-Distribution Robustness (ODR)

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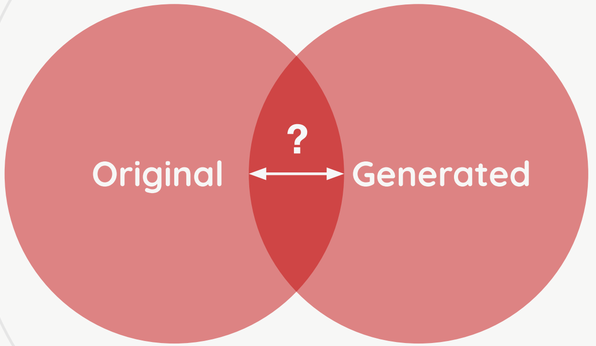
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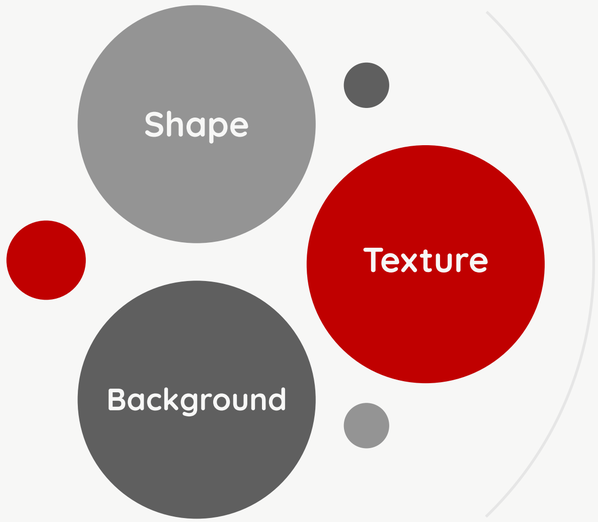
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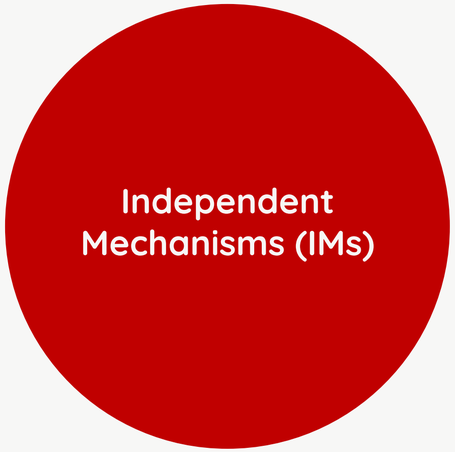
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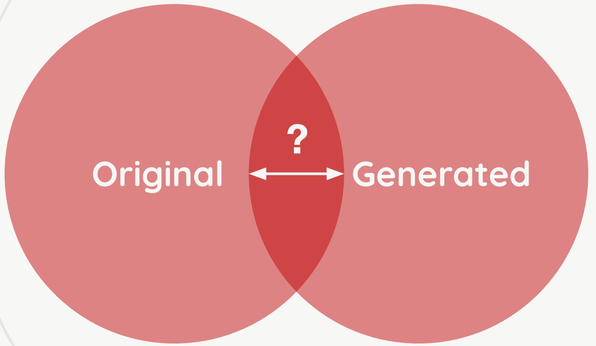
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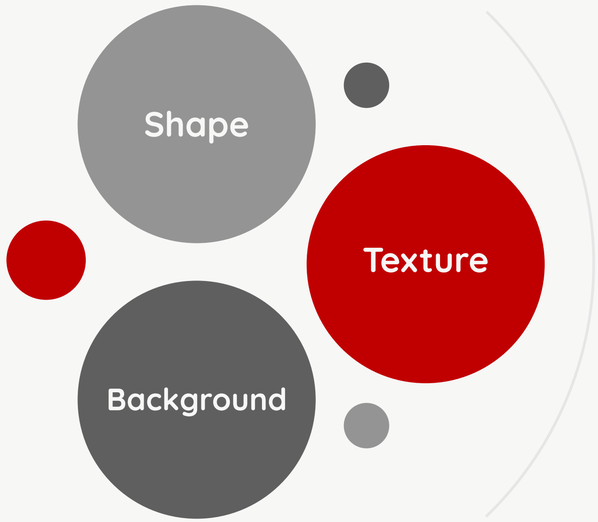
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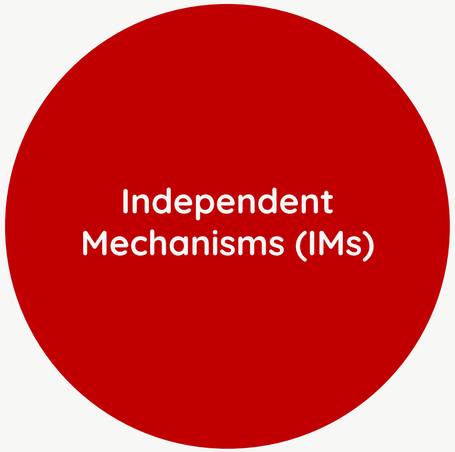
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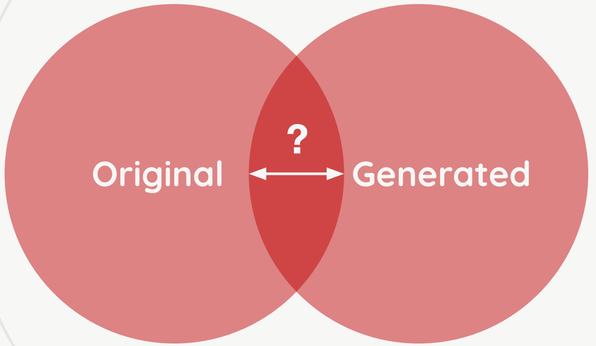
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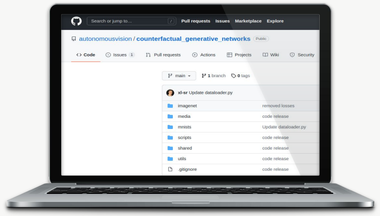


Out-of-Distribution Robustness (ODR)

Methodology

Models

The CGN model is publicly available on GitHub



Datasets

Various variants of MNIST and ImageNet



Experimental Setup + Metrics

Re-implement based on description of paper



Single-Shot Three-Dimensional Reconstruction for Colonoscopic Image Analysis

Computational Requirements

112 + 48 GPU hours on a 1080Ti node (Lisa)

SURF SARA

Experimental results of reproducibility study

Claim 1: High-Quality Counterfactuals (HQC)

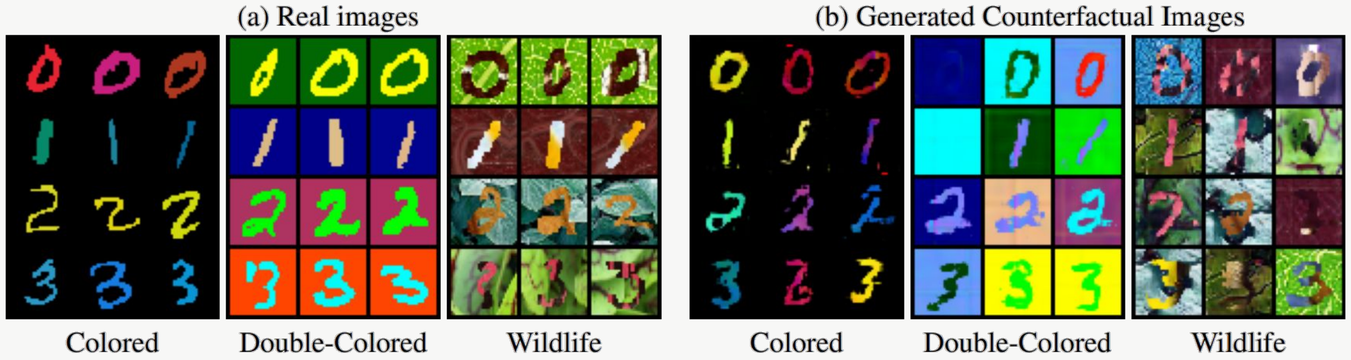


***Figure 2**. Reproduced qualitative results on MNIST variants*

Shape	Racer	Trench coat	Turtle	Vase	Malinois	Barrel
Texture	Clock	Cab	Cauliflower	Elephant	Viper	Piggy bank
Background	Toucan	Coral reef	Mushroom	Alp	Spider	Ibex

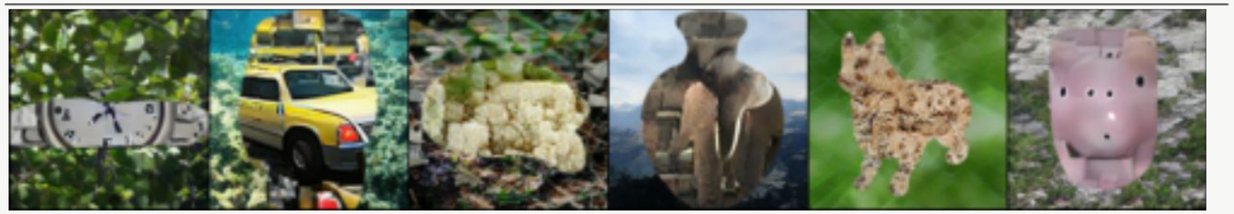


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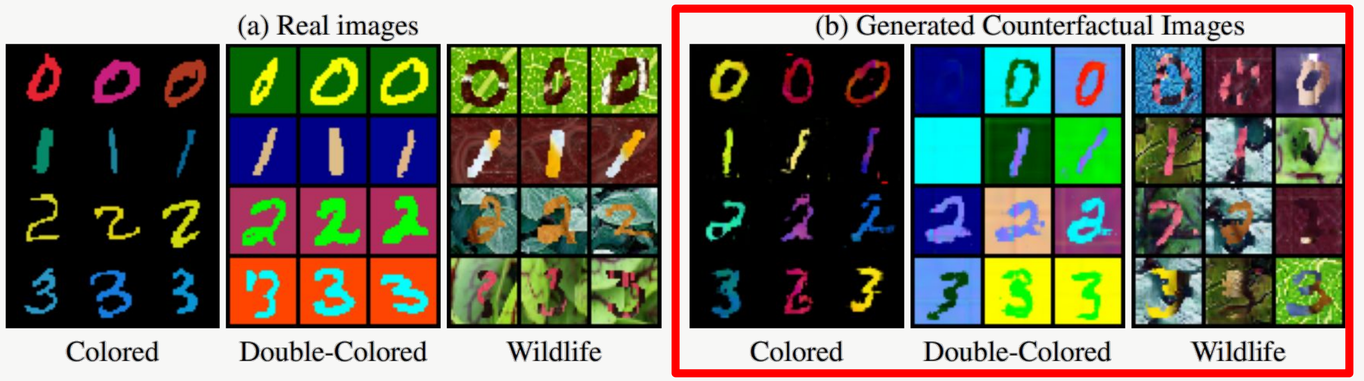


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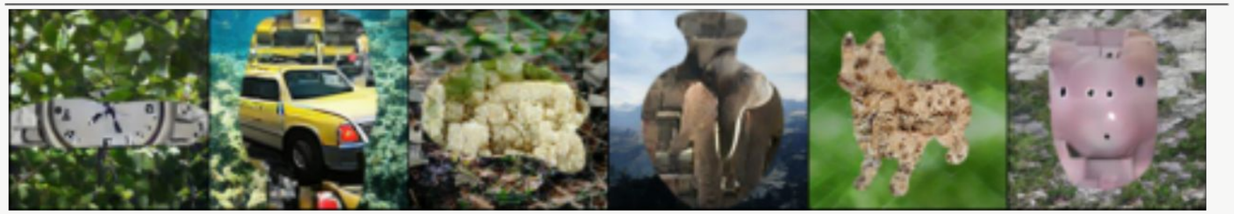


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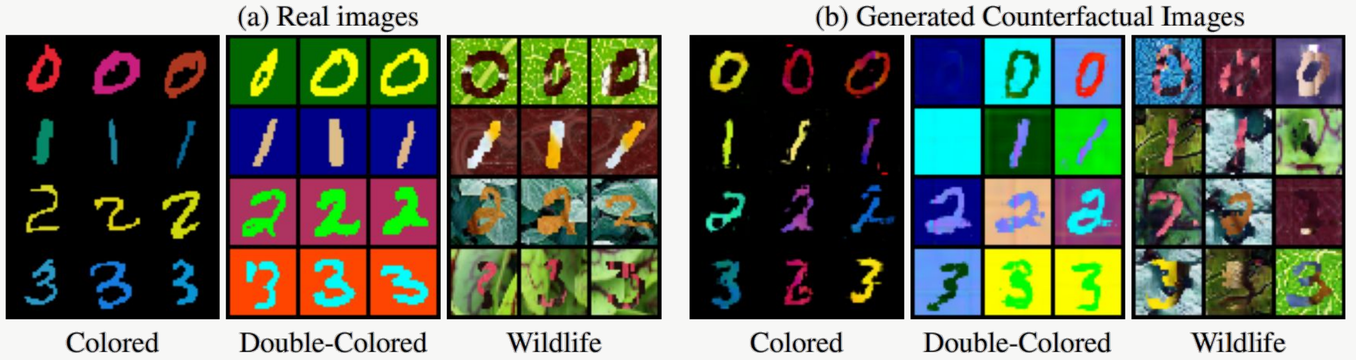


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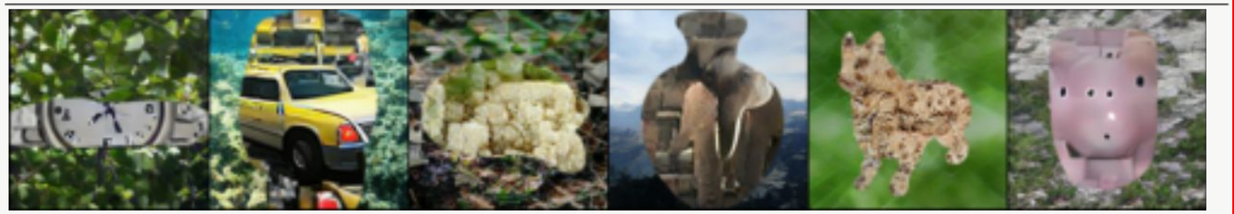
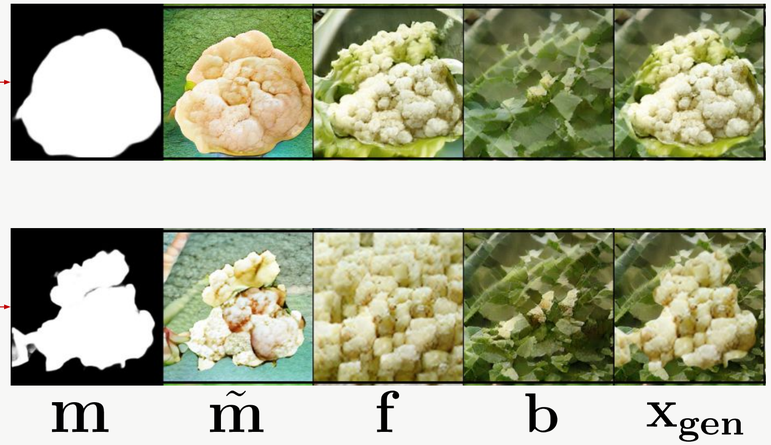


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Claim 2: Inductive Bias Requirements (IBR)

$\mathcal{L}_{shape}$	$\mathcal{L}_{text}$	$\mathcal{L}_{bg}$	$\mathcal{L}_{rec}$	IS ⇑	μ_{mask}
✗	✓	✓	✓	100.8 | 85.9	0.3 | 0.2
✓	✗	✓	✓	186.5 | 198.4	0.4 | 0.9
✓	✓	✗	✓	200.9 | 195.6	0.1 | 0.1
✓	✓	✓	✗	19.3 | 38.4	0.4 | 0.3
✓	✓	✓	✓	156.1 | 130.2	0.3 | 0.3
BigGAN (Upper Bound)				202.9	-

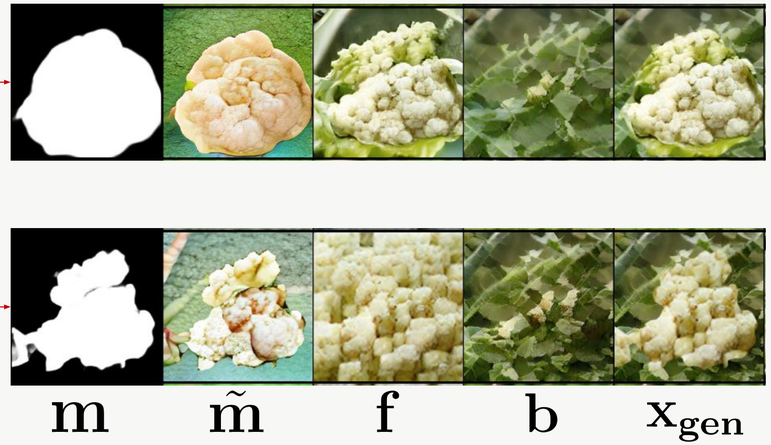
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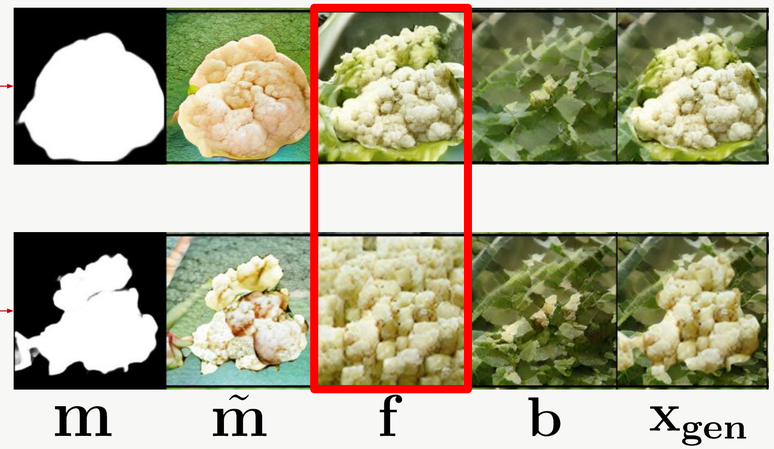
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Claim 3: Out-of-Distribution Robustness (ODR)

***Table 2**. Reproduced qualitative results on MNIST variants.*

Setting	C-MNIST		DC-MNIST		W-MNIST	
	Train ⇑	Test ⇑	Train ⇑	Test ⇑	Train ⇑	Test ⇑
Original	99.7 | 99.5	37.6 | 35.9	100 | 100	10.5 | 10.3	100 | 100	10.8 | 10.1
GAN	99.6 | 99.8	32.5 | 40.7	100 | 100	10.6 | 10.8	99.9 | 100	11.2 | 10.4
CGN	99.4 | 99.7	92.3 | 95.1	94.8 | 97.4	86.5 | 89.0	95.5 | 99.2	81.4 | 85.7
O + GAN	99.6 | 99.8	41.5 | 40.7	100 | 100	10.0 | 10.8	100 | 100	11.1 | 10.4
O + CGN	99.2 | 99.7	95.9 | 95.1	96.9 | 97.4	85.5 | 89.0	96.8 | 99.2	62.8 | 85.7

***Table 3**.Shape biases of independent classifiers*

Trained on	Shape Bias	top-1 ⇑	top-5 ⇑
IN + GCN/Shape	54.8	74.0	91.7
IN + GCN/Text	16.7		
IN + GCN/Bg	22.9		
IN-mini + GCN/Shape	58.8	56.5	79.3
IN-mini + GCN/Text	22.6		
IN-mini + GCN/Bg	24.7		

***Table 4**. Evaluation of robustness against adversarially chosen backgrounds*

Trained on	IN-9 ⇑	Mixed-Same ⇑	Mixed-Rand ⇑	BG-Gap ⇓
IN	95.6	86.2	78.9	7.3
SIN	89.2	73.1	63.7	9.4
IN + SIN	94.7	85.9	78.5	7.4
Mixed-Rand	73.3	71.5	71.3	0.2
IN + CGN	94.2	83.4	80.1	3.3
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CGN	99.4 | 99.7	[illegible] | 95.1	94.8 | 97.4	86.5 | [illegible]	95.5 | 99.2	81.4 | 85.7
O + GAN	99.6 | 99.8	41.5 | 40.7	100 | 100	10.0 | 10.8	100 | 100	11.1 | 10.4
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Results beyond original paper

Explainability analysis: Visualizing features

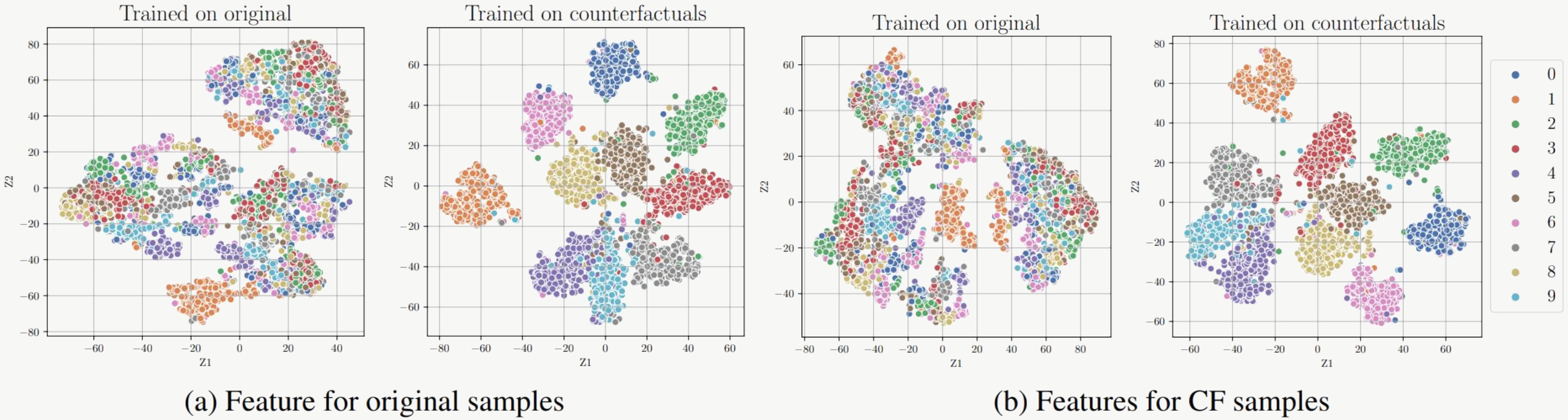


Figure 4. *Feature space visualization of a CNN classifier trained on on colored MNIST variants*

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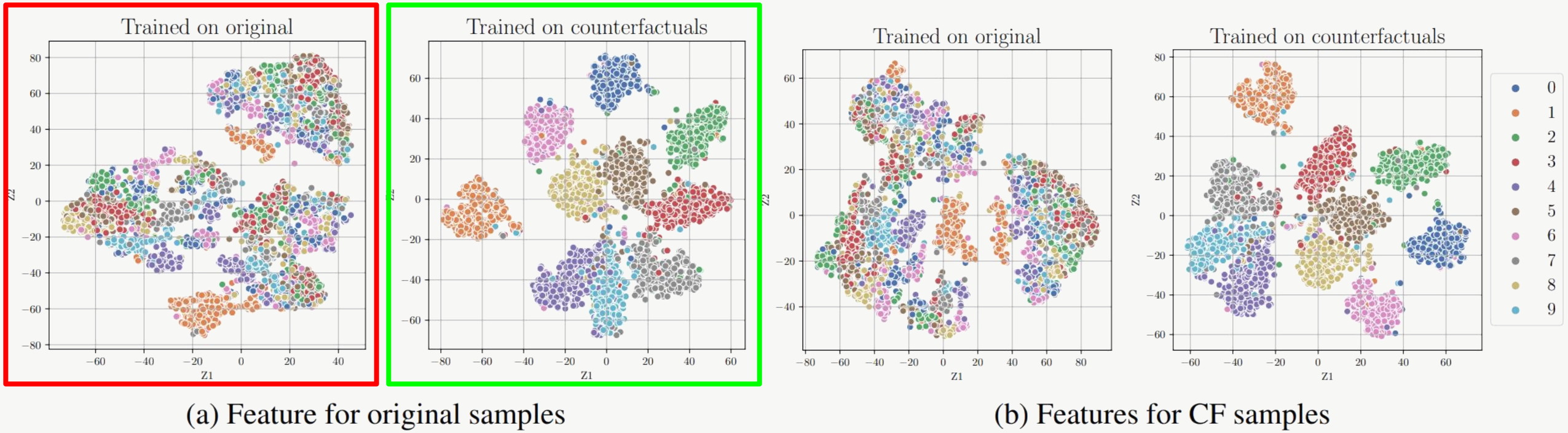


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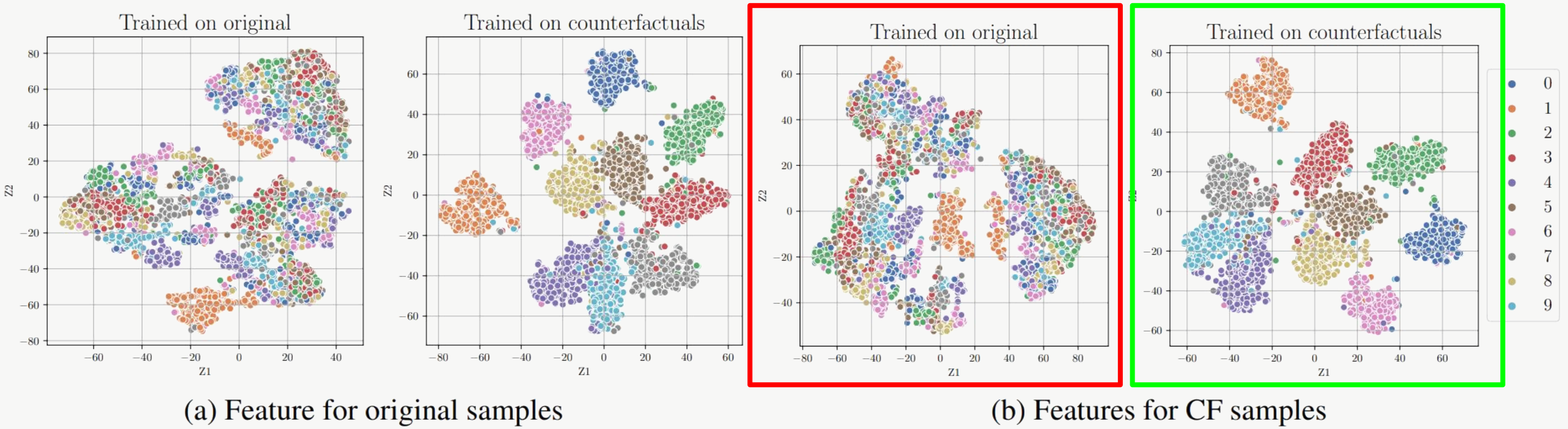


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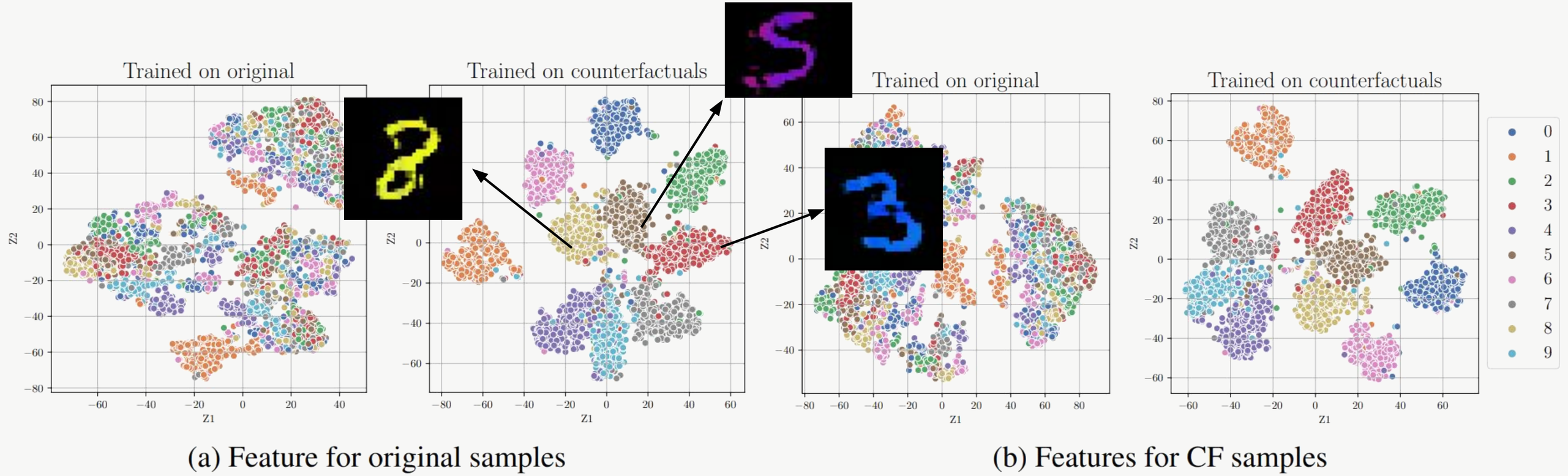


Figure 4. *Feature space visualization of a CNN classifier trained on on colored MNIST variants*

Explainability analysis: What does the model focus on?

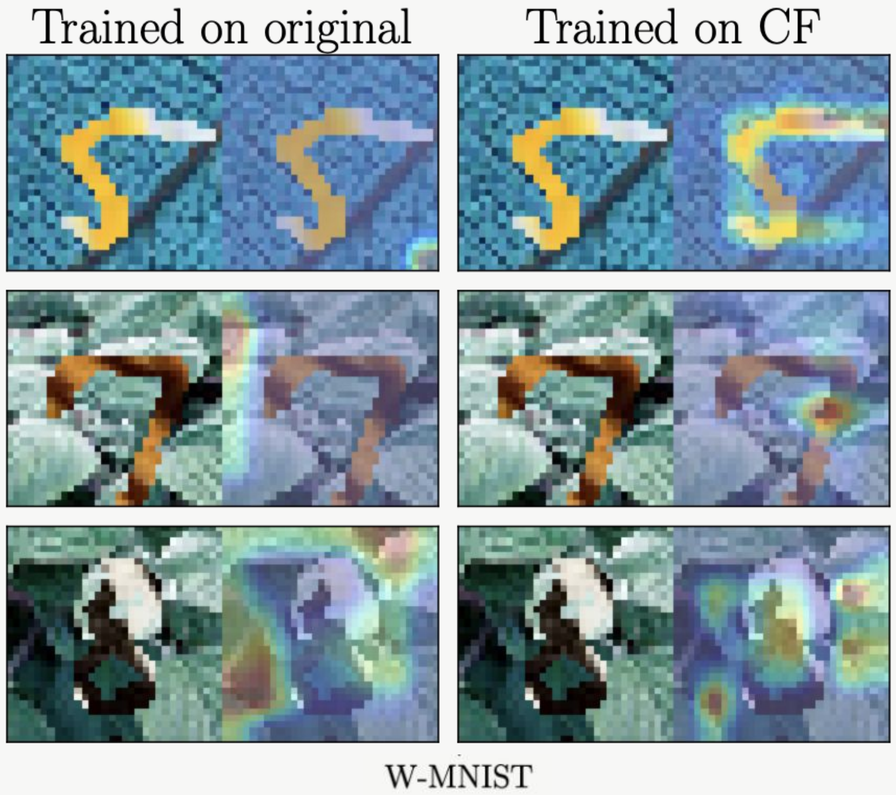


***Figure 5**. GradCAM heatmap visualized on W-MNIST samples*

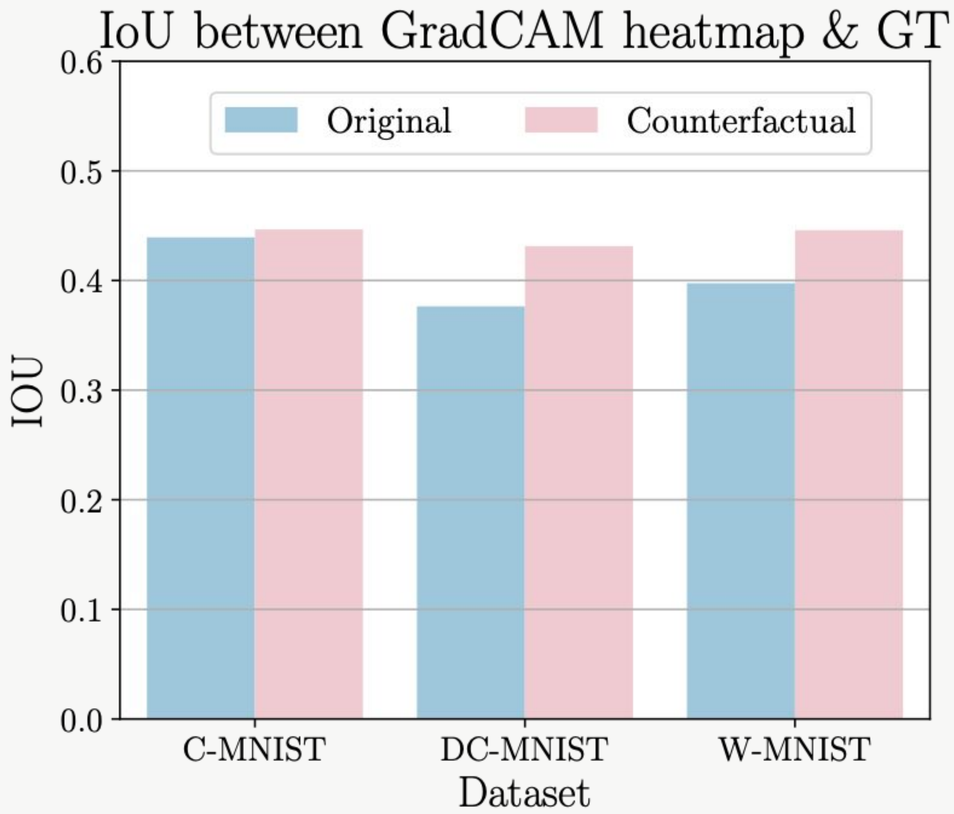


***Figure 6**. Metric to quantify areas where the model focuses on*

[Demo for ImageNet]

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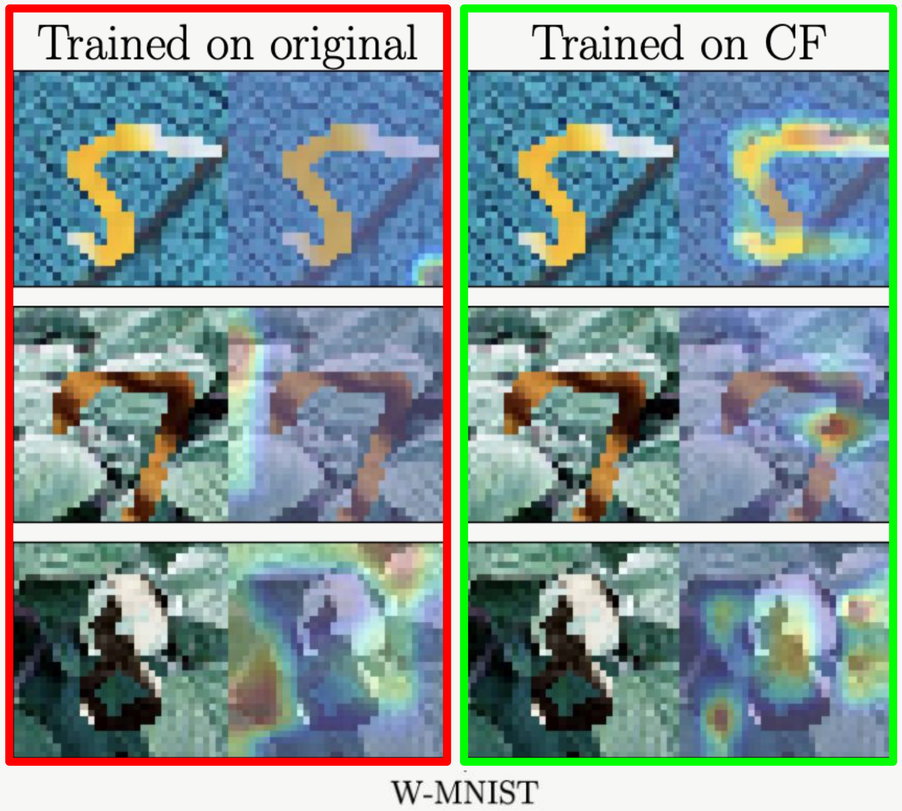


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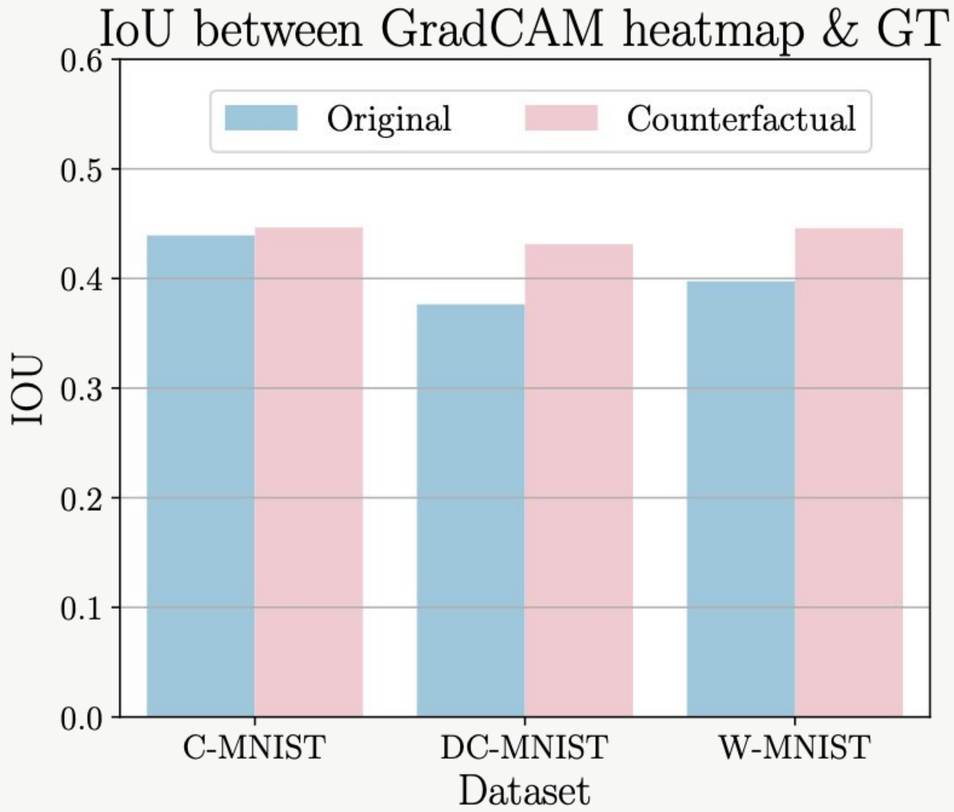


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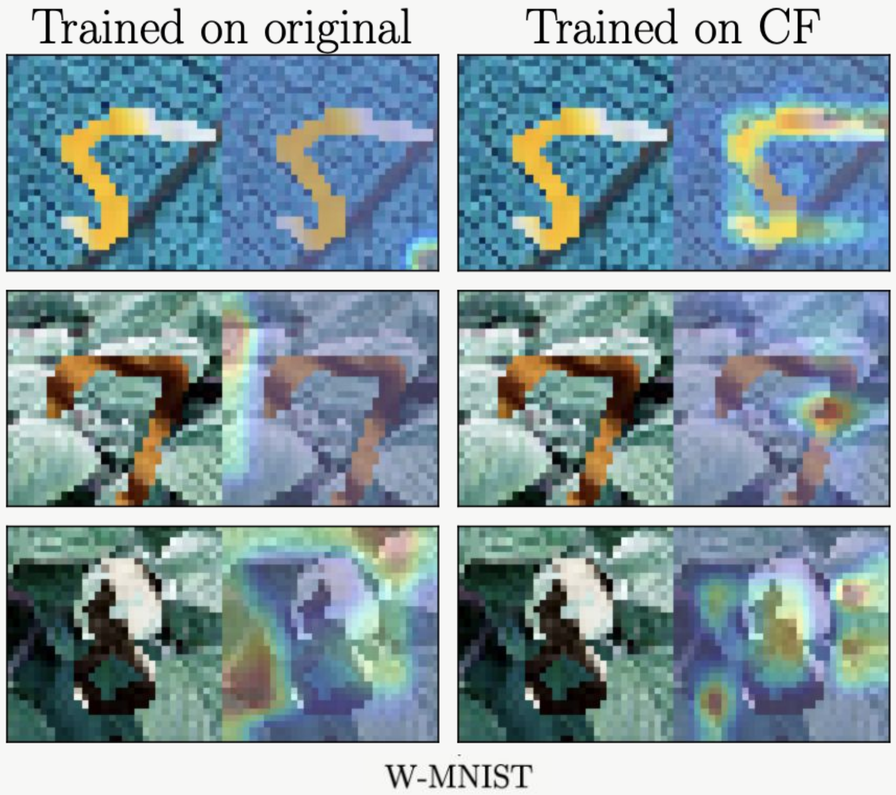


***Figure 5**. GradCAM heatmap visualized on W-MNIST samples*

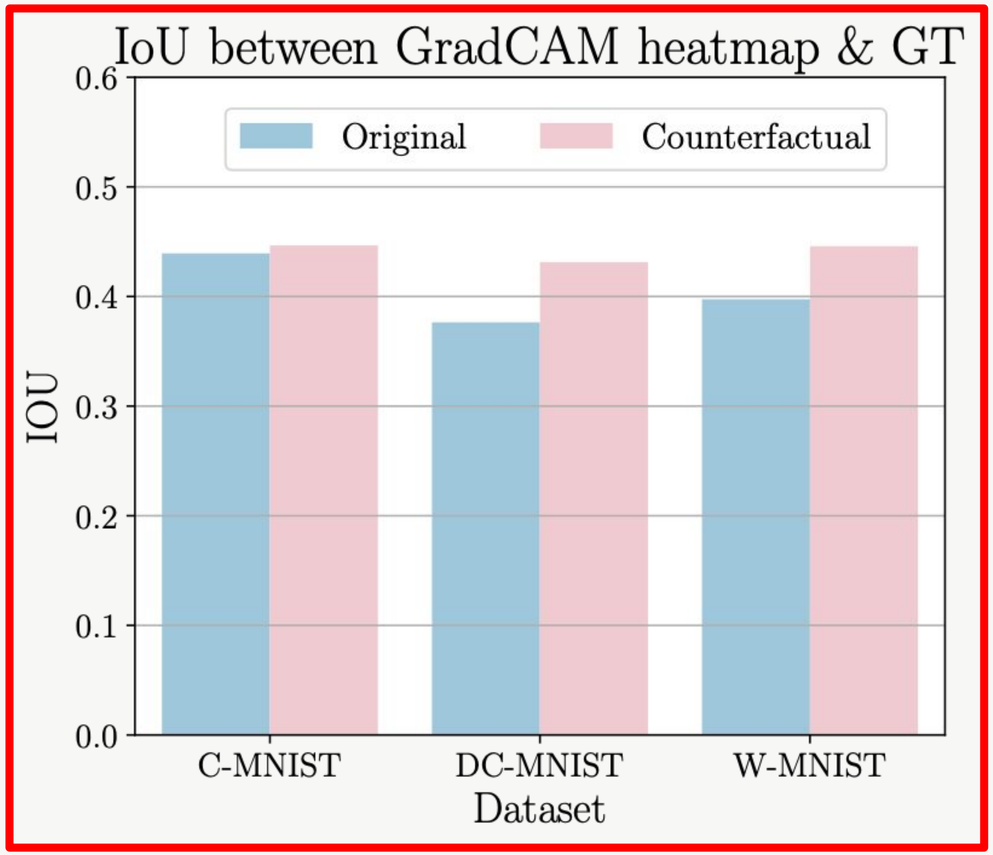


***Figure 6**. Metric to quantify areas where the model focuses on*

[Demo for ImageNet]

Our experience & Lessons Learned

Fairness, Research & Management



Great overview of topics such as fairness, accountability and AI ethics in general!



Great (first) research experience unlike other course assignments!



Managing time/workload under a deadline



Collaborating in a group over a research project

Tips & Suggestions: Research

Several aspects to research (specific to reproducibility)

- Read and understand thoroughly
- Identify key contributions of the paper
- Identify key experiments that support these
- Identify drawbacks and possible extensions
- Coding
- Experimenting
- Writing and presenting
- Submitting
- ...



Ask the right questions!



Qualitative analysis generally helps to get a nice intuition beyond numbers!



Look at reproducibility papers of previous years for inspiration and structure!



Do not fixate on reproducing the exact numbers. Looks for matching trends!

Tips & Suggestions: Management



It helps to appoint a lead for each broad vertical – but the lead should not do everything!



Communication is the key!

- Setup a chat for real-time comms (Discord/WhatsApp)
- It helps to meet regularly (daily - albeit for 15 mins)



Start writing early and not just a day before the deadline! We started in week 1.



Learn from each other!





Q&A

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